

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **2,3,4,5-Tetrafluorobenzoyl chloride**  
 Product Description: **2,3,4,5-Tetrafluorobenzoyl chloride**  
 Cat No. : L18132  
 CAS No 94695-48-4  
 Molecular Formula C7 H Cl F4 O

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                     |
|----------------------------------------------------|---------------------|
| Skin Corrosion/Irritation                          | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation                  | Category 1 (H318)   |
| Specific target organ toxicity - (single exposure) | Category 3 (H335)   |

**Label Elements**


Signal Word

Danger

Hazard Statements

# SAFETY DATA SHEET

2,3,4,5-Tetrafluorobenzoyl chloride

Revision Date 24-Mar-2025

H335 - May cause respiratory irritation  
H314 - Causes severe skin burns and eye damage

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area

### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P310 - Immediately call a POISON CENTER or doctor  
P374 - Fight fire with normal precautions from a reasonable distance  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P235 - Store in a well-ventilated place. Keep cool  
P404 - Store in a closed container

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Combustible liquid  
Lachrymator (substance which increases the flow of tears)  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                           | CAS No     | Weight % |
|-------------------------------------|------------|----------|
| 2,3,4,5-Tetrafluorobenzoyl chloride | 94695-48-4 | 99       |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                |
| <b>Skin Contact</b>                       | Immediate medical attention is required. Wash off immediately with plenty of water for at least 15 minutes.                                      |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Get medical attention.                                                                                                   |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                                                       |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### Most important symptoms and effects, both acute and delayed

# SAFETY DATA SHEET

2,3,4,5-Tetrafluorobenzoyl chloride

Revision Date 24-Mar-2025

Difficulty in breathing. Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Combustible material. Flammable. Containers may explode when heated.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Gaseous hydrogen fluoride (HF), Hydrogen chloride gas.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Use spark-proof tools and explosion-proof equipment. Avoid contact with skin, eyes or clothing. Avoid breathing dust/fume/gas/mist/vapors/spray. Avoid ingestion and inhalation. Remove all sources of ignition. Keep away from open flames, hot surfaces and sources of ignition.

# SAFETY DATA SHEET

2,3,4,5-Tetrafluorobenzoyl chloride

Revision Date 24-Mar-2025

## Conditions for Safe Storage, Including any Incompatibilities

Corrosives area. Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

Particulates filter conforming to EN 143 or Acid gases filter Type E Yellow conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

Colorless

#### **Physical State**

Liquid

#### **Odor**

Strong

#### **Odor Threshold**

No data available

# SAFETY DATA SHEET

2,3,4,5-Tetrafluorobenzoyl chloride

Revision Date 24-Mar-2025

|                     |                          |                                   |
|---------------------|--------------------------|-----------------------------------|
| pH                  | No information available |                                   |
| Melting Point/Range | No data available        |                                   |
| Softening Point     | No data available        |                                   |
| Boiling Point/Range | No information available |                                   |
| Flash Point         | 90 °C / 194 °F           | Method - No information available |

|                          |                   |        |
|--------------------------|-------------------|--------|
| Evaporation Rate         | No data available |        |
| Flammability (solid,gas) | Not applicable    | Liquid |
| Explosion Limits         | No data available |        |

|                              |                          |             |
|------------------------------|--------------------------|-------------|
| Vapor Pressure               | No data available        |             |
| Vapor Density                | No data available        | (Air = 1.0) |
| Specific Gravity / Density   | 1.580                    |             |
| Bulk Density                 | Not applicable           | Liquid      |
| Water Solubility             | No information available |             |
| Solubility in other solvents | No information available |             |

## Partition Coefficient (n-octanol/water)

|                           |                          |                                        |
|---------------------------|--------------------------|----------------------------------------|
| Autoignition Temperature  | No data available        |                                        |
| Decomposition Temperature | No data available        |                                        |
| Viscosity                 | No data available        |                                        |
| Explosive Properties      |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties      | No information available |                                        |

|                   |              |
|-------------------|--------------|
| Molecular Formula | C7 H Cl F4 O |
| Molecular Weight  | 212.53       |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Moisture sensitive. Stable.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

### Conditions to Avoid

Incompatible products. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

# SAFETY DATA SHEET

2,3,4,5-Tetrafluorobenzoyl chloride

Revision Date 24-Mar-2025

Strong oxidizing agents. Strong bases. Alcohols. Amines. Ammonia.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Gaseous hydrogen fluoride (HF). Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

|                                                   |                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Information</b>                        | No acute toxicity information is available for this product                                                                                                                                                                                                                                                                                      |
| <b>(a) acute toxicity;</b>                        |                                                                                                                                                                                                                                                                                                                                                  |
| Oral                                              | No data available                                                                                                                                                                                                                                                                                                                                |
| Dermal                                            | No data available                                                                                                                                                                                                                                                                                                                                |
| Inhalation                                        | No data available                                                                                                                                                                                                                                                                                                                                |
| <b>(b) skin corrosion/irritation;</b>             | Category 1 B                                                                                                                                                                                                                                                                                                                                     |
| <b>(c) serious eye damage/irritation;</b>         | Category 1                                                                                                                                                                                                                                                                                                                                       |
| <b>(d) respiratory or skin sensitization;</b>     |                                                                                                                                                                                                                                                                                                                                                  |
| Respiratory                                       | No data available                                                                                                                                                                                                                                                                                                                                |
| Skin                                              | No data available                                                                                                                                                                                                                                                                                                                                |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                                                                                                                                                                                                                                |
| <b>(f) carcinogenicity;</b>                       | No data available                                                                                                                                                                                                                                                                                                                                |
|                                                   | There are no known carcinogenic chemicals in this product                                                                                                                                                                                                                                                                                        |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                                                                                                                                                                                                                                                                                                |
| <b>(h) STOT-single exposure;</b>                  | Category 3                                                                                                                                                                                                                                                                                                                                       |
| Results / Target organs                           | Respiratory system.                                                                                                                                                                                                                                                                                                                              |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                                                                                                                                                                                                                                |
| Target Organs                                     | No information available.                                                                                                                                                                                                                                                                                                                        |
| <b>(j) aspiration hazard;</b>                     | No data available                                                                                                                                                                                                                                                                                                                                |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                                                                                                                   |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. |

# SAFETY DATA SHEET

2,3,4,5-Tetrafluorobenzoyl chloride

Revision Date 24-Mar-2025

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

**Persistence and degradability** No information available

**Bioaccumulative potential** No information available

**Mobility in soil** No information available.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not flush to sewer Large amounts will affect pH and harm aquatic organisms

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1760  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive liquid, n.o.s. (2,3,4,5-TETRAFLUOROBENZOYL CHLORIDE)

### Road and Rail Transport

UN-No UN1760  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive liquid, n.o.s. (2,3,4,5-TETRAFLUOROBENZOYL CHLORIDE)

### IATA

UN-No UN1760  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive liquid, n.o.s. (2,3,4,5-TETRAFLUOROBENZOYL CHLORIDE)

ALFAAL18132

# SAFETY DATA SHEET

2,3,4,5-Tetrafluorobenzoyl chloride

Revision Date 24-Mar-2025

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories** X = listed

| Component                           | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL |
|-------------------------------------|--------|------|-----|-------|------|------|-------|------|------|
| 2,3,4,5-Tetrafluorobenzoyl chloride | -      | -    | -   | -     | -    |      | X     | -    | -    |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By** Health, Safety and Environmental Department  
**Revision Date** 24-Mar-2025  
**Revision Summary** Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health**



# SAFETY DATA SHEET

2,3,4,5-Tetrafluorobenzoyl chloride

Revision Date 24-Mar-2025

---

## (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**